



AMD Strix Halo FP11 Engineering Interlock

CUSTOMER ENGINEERING
FEBRUARY 2025

AMD CONFIDENTIAL – NDA REQUIRED

ing@mapleleaf-info.com 2025-11-24 Strix Halo FP11 Engineering Interlock February 2025 - for Standard Customer

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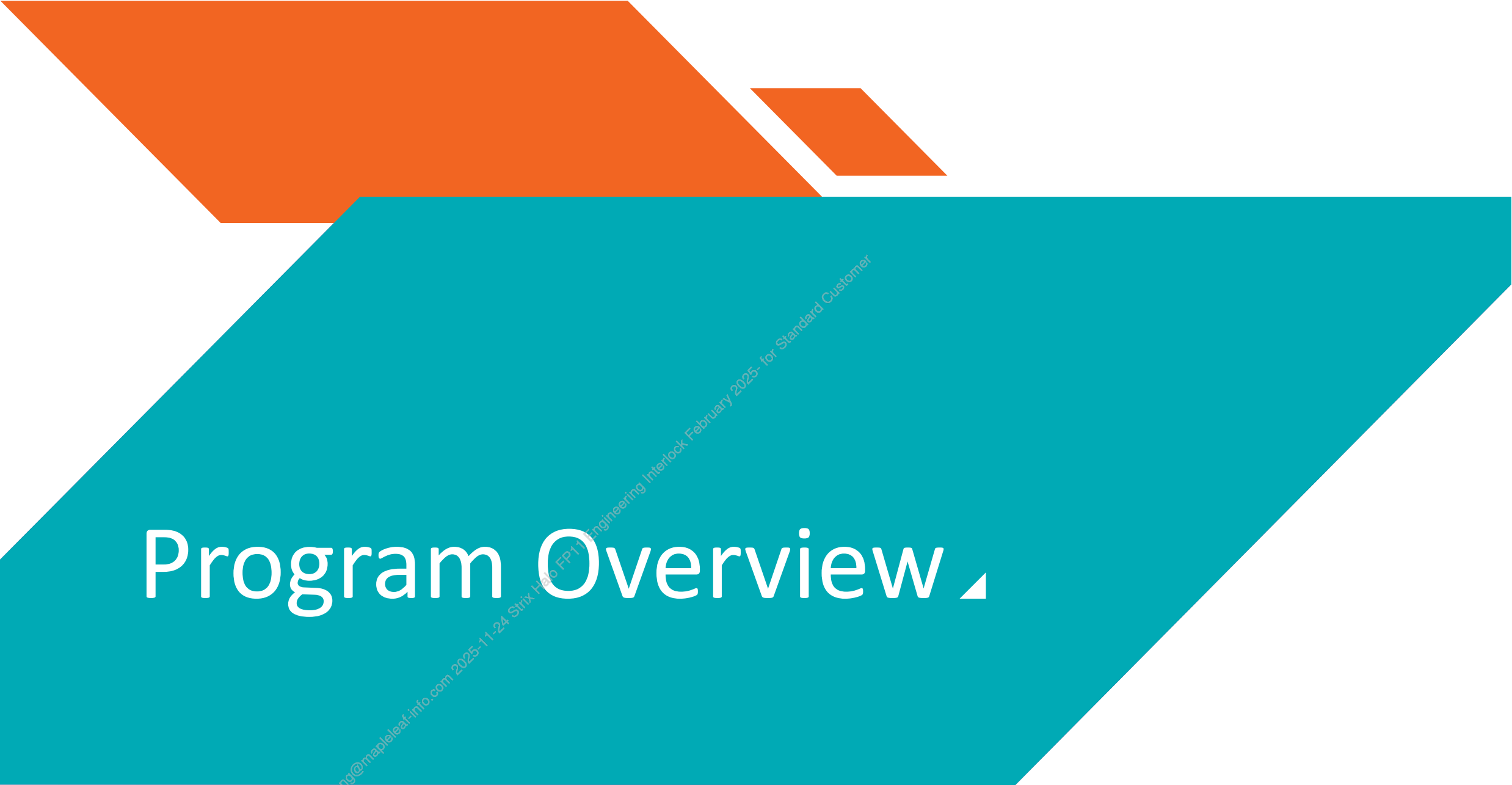


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The slide features a large teal-colored area at the bottom and two orange-colored shapes at the top left. One is a large parallelogram, and the other is a smaller, tilted rectangle. The text "Program Overview" is centered in the teal area.

Program Overview

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STRIX HALO FAMILY OVERVIEW

- ▲ Strix HALO APU (Accelerated Processing Unit) is also known as
 - AMD Family 1Ah Models 70h-77h Processor
- ▲ Zen5: Core name of the CPU architecture in Strix Halo FP11 APUs
- ▲ Radeon Navi3.5: Name of the Graphics core in Strix Halo FP11 APUs
- ▲ Strix Halo platform includes Hardware Security Processor (HSP) in addition to Platform Security Processor (PSP)
- ▲ Maple APU-based customer reference board (CRB) based on LPDDR5x with all the required debug and feature test capability

APU / Platform Overview

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CPU CORE & PACKAGE

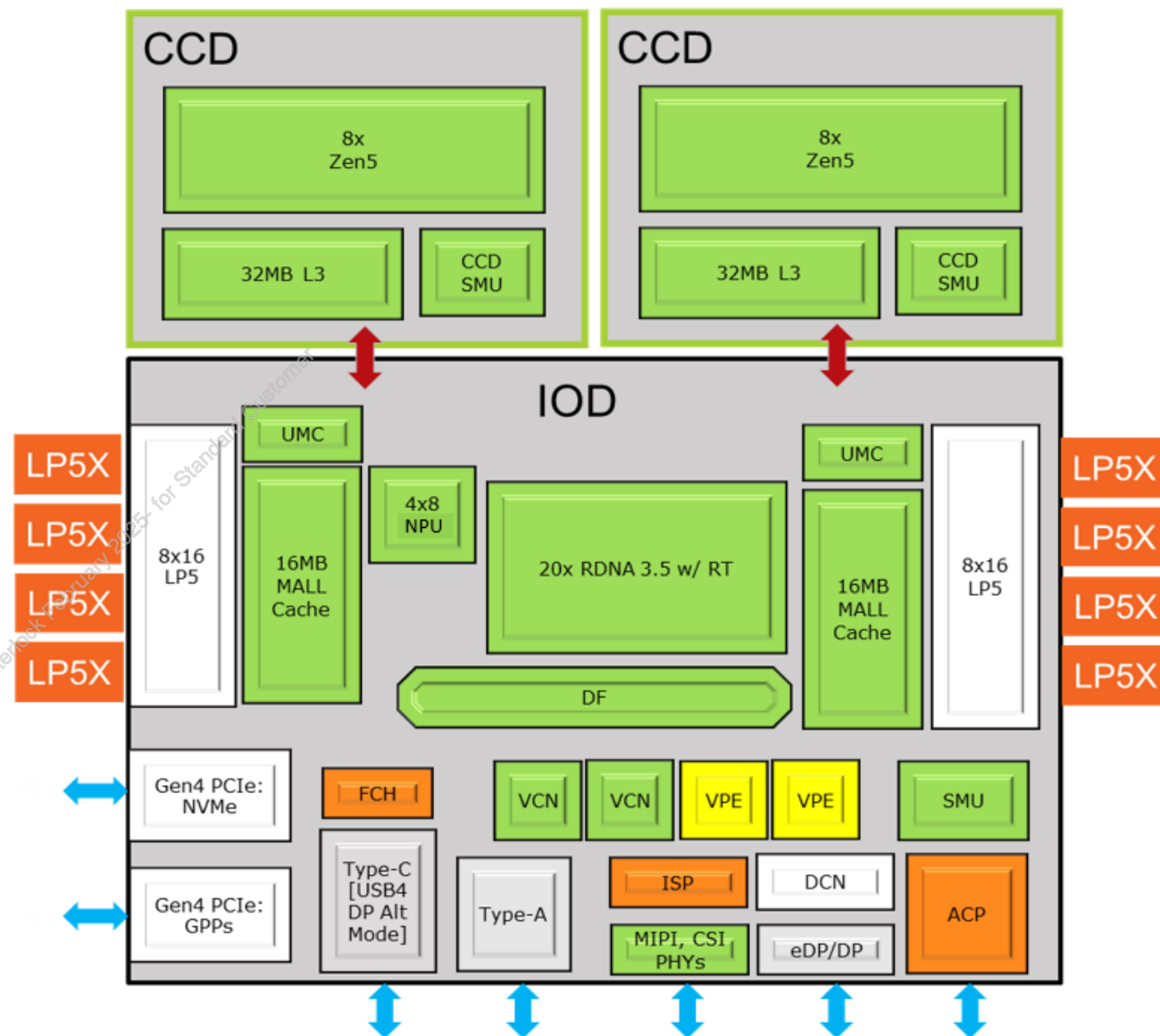


	Strix Halo
CPU Architecture	Zen5
CPU Cores	16 Cores; 2 CCDs of 8 each
L1 Cache	L1 Instruction Cache = 32 KB, L1 Data Cache = 32 KB (per core)
L2 Cache	L2 Cache = 1 MB per core
L3 Cache	64MB; 32 MB per CCD
Graphics	Navi 3.5 graphics with up to 20 WGP (Work Group processors)
MALL Cache	32MB
Security Processor	Microsoft Security processor to provide fTPM 2.0 and additional security features
Package Type	Lidless BGA, uniform 0.8mm ball pitch; FP11
Package Size	37.5mm x 45mm
Max Z-height	1.7 mm
Ball Count	2077

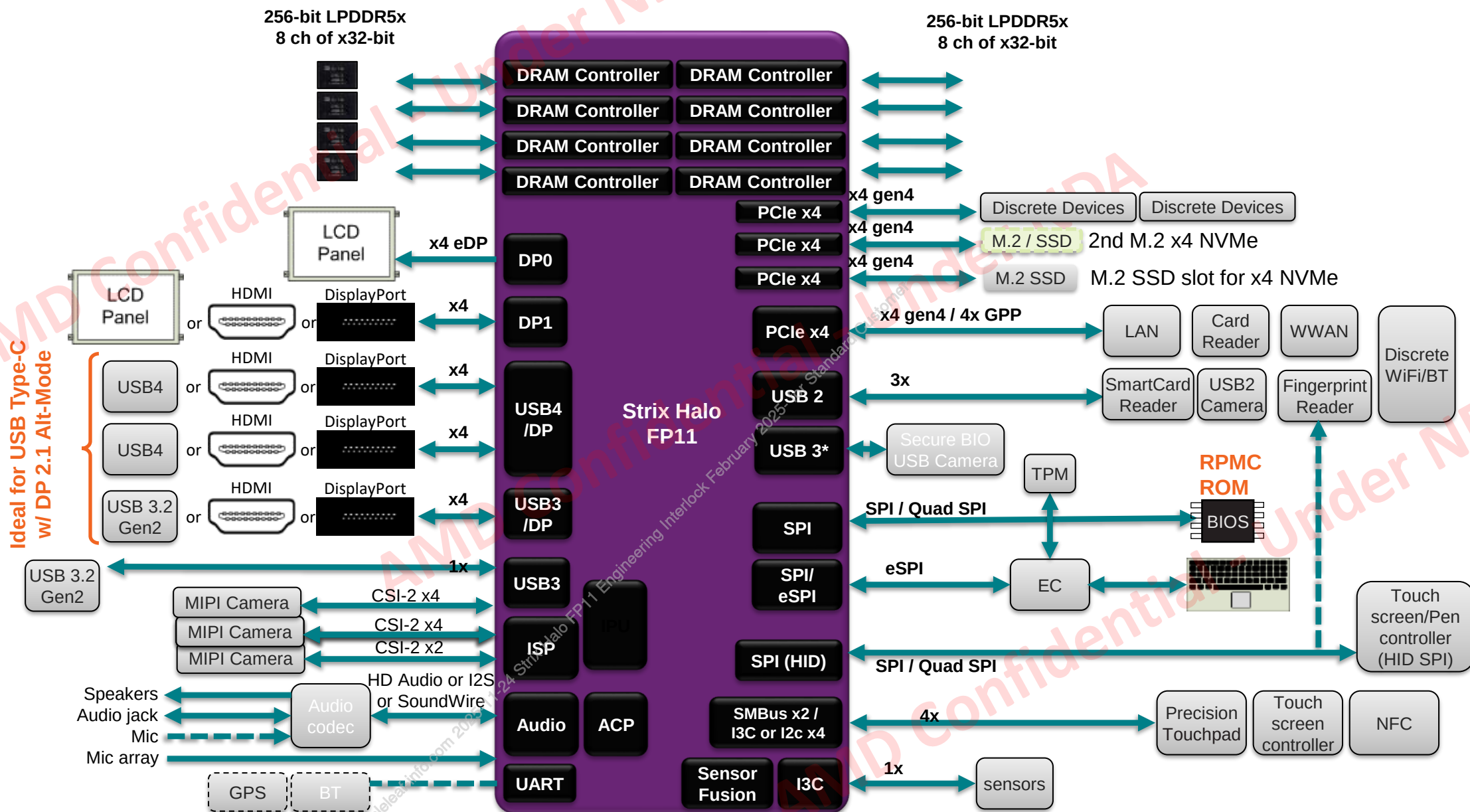


- Competes with Mx Pro designs in Consumer/Creator/Pro/Mobile Workstation
- Targeted to exceed Gaming/MWS performance vs Nvidia 4070/4060 discrete GPUs at the same power level.

- AMD Chiplet Technology using advanced packaging technology
- CPU: Zen5 + 32MB L3 in separate CCDs
- GPU: RDNA3.5+ with 20WGP, 32MB MALL Cache, Power Management
- Memory: 256b LP5X 8000 MT/s
- Video: Upgraded VCN4.0



STRIX HALO FP11 SYSTEM BLOCK DIAGRAM



APU & PLATFORM



	Strix Halo
CPU Architecture	Zen5, Up to 16 Cores
Family	Family 1Ah Models 70h-77h
Process Technology	N4
Package / TDP	55W nominal (cTDP 45W-120W)
Primary Power Requirements	Requires 3 Primary Voltages (VDDCR, VDDCR_CCD, VDDCR_SOC)
GPU	Radeon Navi3.5, Up to 20 WGP(40 CUs)
Minimum PCB Layers	12L ,Non-HDI, uVIA PCB with LPDDR5x DRAM Down
Reference Board	Maple FP11
dGPU Pairing	Not Supported
ROM Size for BIOS	32 MB/64 MB
Software	Strix Halo PI-FP11, AMD Drivers

FCH & I/O



	Strix Halo
Memory	LPDDR5x 256b (8x 32b Channels), DRAM Down, 8000 MT/s
Discrete Graphics Port	Not Supported
GPP	PCIe Gen4 Up to 16 lanes and up to 6 ports
SATA	No support
Display outputs	5 Ports (eDP, DP, USB4, HDMI, DVI) (3 muxed with USB for USB-C™ with DP Alt. Mode or USB4 or USB3.2) DP2.1 Alt. Mode requires USB-Power Delivery controller separate from SOC
USB4 (40 Gbps)	2 USB4 (40 Gbps)
USB3.2 Gen2 (10Gbps)	1 USB 3.2 Gen2 (10 Gbps) for USB-C™ with DP Alt. Mode port 1 USB 3.2 Gen2 (10 Gbps) for Type A port
USB2.0	3 USB 2.0 Ports (shared controller with USB 3.2 ports)
USB3.2 Gen2 (10Gbps)	1 USB 3.2 Port (dedicated controller) for Secure BIO
MIPI bus	10 lanes, x4 + x4 + x2
SMBus	2 Ports Shared with I ² C
LPC	LPC is removed and no longer supported
SPI	Yes
eSPI	Yes
UART	4 Ports: 2 4-wire or 4 2-wire
I ² C / I3C	4 Ports
Clock Generator	Yes
Sensor Fusion Hub	Yes (1 I ² C, 6 GPIOs or 1 I ² C, 1 I3C, 4 GPIOs)
Miracast Wireless Display	Yes

USB PORT SUPPORT IN DETAILS



Port type	Strix Halo	Number of Ports
USB-C with DP Alt-mode	USB4 40 Gb/s (20 Gb/s per lane) UHBR20 DP2.1 20 Gb/s Alt Mode USB3.2 Gen2 10Gb/s	2
USB-C with DP Alt-mode	USB3.2 Gen2 10Gb/s DP2.1 10 Gb/s Alt Mode	1
USB-A	USB3.2 Gen2 10Gb/s	1
USB2	Shared controller	3
USB3.2	Secure BIO – Dedicated controller	1

STRIX HALO NOTEBOOK PLATFORMS MEMORY CONFIGURATION



	Strix Halo FP11 - LP5
Technology	LPDDR5x 1.05V/0.5V
Width	256
Layers	12L, must have Micro Via
DPC	MD
Frequency	8000 MT/s
Min Freq	1600
Error Correction	Link ECC
Form Factor	MD
1up/1down	NA
DRAM PKG Width	LPDDR5x x32
Max Capacity	128 GB (16Gb/die,128Gb/pkg)
Power management	DVFS: VDD_MEM (PHY Power) is supported DVFS: VDD2H and VDD2L (DRAM Power) is supported DVFSQ is not supported eDVFS (enhanced DVFS) is not supported

Two orange geometric shapes: a large parallelogram and a smaller parallelogram to its right, both with a slight 3D effect.

Silicon Sample Usage Guidance

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STRIX HALO SAMPLE/CUSTOMER TEST ALIGNMENT GUIDANCE



Test Plan Recommendation

▪ **OEM A-Test:**

FP11 - Start test with ES samples and 0.7.0 SW stack

Functionality: Basic I/O Functionality, Win BOOT, limited thermal calibration, limited OPNs

Usage: Limited Power cycling/GPU stress, Win BOOT, Thermal design, BIOS development, electrical, system bring up & debug

Performance: ES = 50% performance/battery life optimization

▪ **OEM B-Test:**

FP11 - Start test with PC samples and 0.8.0 SW stack

Functionality: Full I/O Functionality, power cycling, speed & power bin OPNs

Usage: Battery Life testing, Performance Benchmarking, Graphic & CPU Stress Testing, Signal Integrity, Reliability Testing, Regulatory Cert testing, OS Product Feature Testing, Application base Testing, Hardware Config testing

Performance: PC: 80% performance/battery life optimization

▪ **OEM C-Test:**

FP11 - Start Test with PR samples and 1.0.0 SW stack

Functionality: No fuse change for silicon (final settings, final GPU drivers, etc.)

Usage: All testing; Regression and & OS Regulatory Certification testing

Performance: PR: 100% performance/battery life optimization

Feature Enablement

Definitions:

Enable = feature is working but may not be 100% bug free

Optimized = feature has been optimized and no more changes are being made

STRIX HALO SECURITY FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
AMD Memory Guard	Disable	Enable	Enable	
Intrusion detection	Enable	Enable	Enable	
A/B Partition-based Firmware Update and Recovery	Enable	Enable	Enable	
AMD Secure Parts	Disable	Enable	Enable	
AMD Shadow Stack	Enable	Enable	Enable	
Bitlocker	Enable	Enable	Enable	
Discrete TPM (dTPM)	Enable	Enable	Enable	
Drive Lock	Enable	Enable	Enable	
MSFT Windows Hello	Enable	Enable	Enable	
Memory Access Protection	Disable	Enable	Enable	
MS Application Guard: Device Credential Guard	Enable	Enable	Enable	

STRIX HALO SECURITY FEATURES (CONTINUED)



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
MSFT Windows -System Guard - SMM Protection - SMM Paging Protection	Disable	Enable	Enable	
Platform Secure Boot (PSB)	Disable	Enable	Enable	
PlayReady SL3000 Hardware DRM Support	Disable	Enable	Enable	
Pluton fTPM	Disable	Enable	Enable	
PSP Firmware Anti-Rollback (FAR)	Disable	Disable*	Enable	*Issue listed in 080 Advisory. Fixed in 1.0.0
Replay protection Monotonic Counter (RPMC) - fTPM Support	Disable	Enable	Enable	
ROM Armor v3	Enable	Enable	Enable	
Secure BIO using MIPI Camera	Enable	Enable	Enable	
Secure BIO Using USB	Enable	Enable	Enable	
Virtualization Based Security (VBS)	Enable	Enable	Enable	
Microsoft Secure Core PC (SCPC)	Disable	Enable	Enable	

STRIX HALO MULTIMEDIA FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
HDA Offload(Dual LPAP)	Enable	Enable	Enable	
USB Offload(LPAP)	Enable	Enable	Enable	
Soundwire Offload(LPAP)	Enable	Enable	Enable	
Hardware keyword spotter with customizable Third-party keyword	Enable	Enable	Enable	No 3 rd party support
Bluetooth A2DP Offload (LPAP)	Disable	Enable	Enable	3 rd party dependency
Bluetooth LE Offload (LPAP)	Disable	Enable	Enable	
AMD AI-NR Offload on ACP (HiFi5 + NNE110) for PDM/BT/USB	Disable	Enable	Enable	
Realtek AI-NR Offload on ACP (HiFi5 + NNE110)) for PDM/BT/USB	Disable	Enable	Enable	

Note :

To support > 2 Low Power Audio devices, a RGM (Resource Group Management) supported driver from audio codec IHV is required.

STRIX HALO MULTIMEDIA FEATURES (CONTINUED)



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
AMD Ultrasound Presence Detection(HDA/Soundwire+ S0)	Disable	Enable	Enable	
AMD Ultrasound Presence Detection in Standby(Soundwire+S0i3)	Disable	Enable	Enable	
Elliptic Ultrasound Presence Detection(HDA/Soundwire+ S0)	Disable	Enable	Enable	
Elliptic Ultrasound Presence Detection in Standby(Soundwire+S0i3)	Disable	Enable	Enable	
Encode format Converter (EFC) (tone map, color gamut)	Enable	Enable	Enable	
Miracast Wireless Display	Disable	Enable	Enable	

Note :

To support > 2 Low Power Audio devices, a RGM (Resource Group Management) supported driver from audio codec IHV is required.

STRIX HALO GRAPHICS & DISPLAY



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Display Port 2.0 (HBR3 - 1.4)	Enable	Enable	Enable	
HDMI TMDS	Enable	Enable	Enable	
Display Hot Plug	Enable	Enable	Enable	
Embedded Display Port (eDP) Panel	Enable	Enable	Enable	
DP to HDMI2.1 PCON	Enable	Enable	Enable	
USB-C Alt Mode for Display Port	Enable	Enable	Enable	
DP Multi-Stream Transport (MST)	Enable	Enable	Enable	
ABM (Adaptive Backlight Modulation) 2.4	Enable	Enable	Enable	
Panel Self Refresh (PSR)	Enable	Enable	Enable	
USB4 Display Support	Enable	Enable	Enable	
DPMS (Display Management Signaling)	Enable	Enable	Enable	
Freesync Replay	Enable	Enable	Enable	
eDP ASSR UHBR10 scrambling support	Disable	Disable*	Disable*	*Post 1.0.0 Proof of Concept
Display Port 2.0 (UHBR10)	Disable	Enable*	Enable	*Issue listed in 080 Advisory. To be fixed in 1.0.0

STRIX HALO GRAPHICS & DISPLAY (CONTINUED)



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
UHBR10 for eDP	Disable	Disable*	Disable*	Post 1.0.0 . Depends on production panel availability
UHBR20 for USB4 (DP Alt)	Enable	Enable	Enable	
Brightness 3 on SDR panel	Enable	Enable	Enable	
Display Dynamic Contrast	Enable	Enable	Enable	
Display Stream Compression (DSC)	Enable	Enable	Enable	
Dynamic Refresh Rate Control (v2)	Enable	Enable	Enable	
Eyefinity	Enable	Enable	Enable	
Freesync	Enable	Enable	Enable	
FreeSync Premium Pro (PQ HDR)	Enable	Enable	Enable	
Hardware Flip Queue	Enable	Enable	Enable	
HDCP 2.3(4K Netflix)	Disable	Enable	Enable	
HDMI 2.1 FRL (Fixed Rate Link)	Enable	Enable	Enable	

STRIX HALO GRAPHICS & DISPLAY (CONTINUED)



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
HDMI 2.1 VRR (Variable Refresh Rate)	Enable	Enable	Enable	
High Dynamic Range 10 bit (HDR10)	Enable	Enable	Enable	
Multi-plane overlay (MPO)	Enable	Enable	Enable	
Panel Calibration	Enable	Enable	Enable	
Seamless Boot	Enable	Enable	Enable	
Simultaneous Display (up to 4 Displays)	Enable	Enable	Enable	
VariBright 2.0	Enable	Enable	Enable	
OLED Edge Dimming	Enable	Enable	Enable	
Varibright xLED - Content-Adaptive Contrast Profile (CACP)	Enable	Enable	Enable	
Varibright xLED - Smart Color Profile (SCP)	Enable	Enable	Enable	
OLED Power-Saving Feature	Enable	Enable	Enable	Smart Color Profile (SCP) covers this function.
Seamless Pixel Rate Switch (SPRS)	Enable	Enable	Enable	

STRIX HALO POWER/PERFORMANCE FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Dynamic Voltage and Frequency Scaling (DVFS) (Variable SOC VDD_MEM)	Disable	Enable	Enable	
Windows Mixed Reality	Enable	Enable	Enable	No longer in Windows
Advanced Platform Management Link (APML)	Enable	Enable	Enable	
Collaborative Processor Performance Control (CPPC)	Disable	Enable	Enable	
CPU Power Management	Enable	Enable	Enable	
cTDP - Configurable Thermal Design Power	Enable	Enable	Enable	
System PCC	Disable	Enable	Enable	
Dynamic Power Thermal Configurations (DPTC)	Enable	Enable	Enable	
Fan Speed Control	Enable	Enable	Enable	
Graphics Power Management	Enable	Enable	Enable	

STRIX HALO POWER/PERFORMANCE FEATURES (CONTINUED)



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
On-die Temperature sensing Calibration	Enable	Enable	Enable	
Platform Management Framework (PMF)including Smart PC/Adaptive PC	Disable	Enable	Enable	
Power Delivery (SVI3)	Enable	Enable	Enable	
SDF P-State	Enable	Enable	Enable	
Skin Temperature Aware Power Management (STAPM)	Enable	Enable	Enable	
System Temperature Tracking (STT2.0)	Enable	Enable	Enable	
Thermtrip	Enable	Enable	Enable	
Vari-bright power slider support	Enable	Enable	Enable	

STRIX HALO SENSOR FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Accelerometer Sensor	Enable	Enable	Enable	
Ambient Light Sensor (ALS)	Enable	Enable	Enable	
Gyroscope Sensor	Enable	Enable	Enable	
Magnetometer Sensor	Enable	Enable	Enable	
Navigation Compass Sensor	Enable	Enable	Enable	
Proximity Detection/User Presence Sensor	Enable	Enable	Enable	
Secondary Accelerometer Sensor	Enable	Enable	Enable	
Tablet/Book Mode Detection Sensor	Enable	Enable	Enable	

STRIX HALO IMAGE PROCESSING / ACCELERATION FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
NPU: Full TOPS Enabled	Enable	Enable	Enable	
NPU: MCDM Driver	Disable	Enable	Enable	

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STRIX HALO ISP FEATURES



Exposed by Windows built-in APP (Inbox Camera App, Camera control panel, Windows Hello, Teams)

System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Basic use case – Preview/Video record/Still image capture	Enable	Optimized	Optimized	
Auto exposure/Auto white balance/Manual white balance/ROI for auto exposure and auto white balance	Enable	Optimized	Optimized	
Camera settings manual control – ISO/Saturation/Contrast/Brightness/Sharpness	Enable	Optimized	Optimized	
Digital zoom	Enable	Optimized	Optimized	
Anti-banding/Flick frequency detection	Enable	Optimized	Optimized	
Variable frame rate	Enable	Optimized	Optimized	
Windows Hello	Enable	Optimized	Optimized	
Secure Biometrics(ESS)	Enable	Optimized	Optimized	
Video HDR	Enable	Optimized	Optimized	

STRIX HALO SYSTEM FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
AMD Wireless Button Driver	Enable	Enable	Enable	
ASIC Reset APU	Enable	Enable	Enable	
Cold Boot (S5)	Enable	Enable	Enable	
eSPI	Enable	Enable	Enable	
Hibernate (S4)	Enable	Enable	Enable	
I2c	Enable	Enable	Enable	
I3c	Disable	Disable	Disable	Post 1.0.0
ISP	Enable	Enable	Enable	
Modern Standby (S0i3)	Enable	Enable	Enable	
OS Reset	Enable	Enable	Enable	

STRIX HALO USB FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Detachable Graphics through USB4	Enable	Enable	Enable	
Individual external USB disablement	Enable	Enable	Enable	
TBT3.0 compatibility for USB4 ports	Enable	Enable	Enable	
USB 2.0	Enable	Enable	Enable	
USB 3.2 Gen 1 (5Gbps)	Enable	Enable	Enable	
USB 3.2 Gen 2 (10Gbps)	Enable	Enable	Enable	
USB 4.0	Enable	Enable	Enable	
USB Run time D3	Enable	Enable	Enable	
USB Selective suspend resume	Enable	Enable	Enable	

STRIX HALO SYSTEM FEATURES (CONTINUED)



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
PCIe (x1,& x4) Gen 1 to Gen 4	Enable	Enable	Enable	
SD Card 7.0 (External)	Enable	Enable	Enable	
SPI Mode (Dual, Quad) for all 3 SPI ports	Enable	Enable	Enable	
UART (Universal Asynchronous Receiver Transmitter)	Enable	Enable	Enable	
WLAN/WWAN (External)	Enable	Enable	Enable	
Variable Graphics Memory	Disable	Enable	Enable	Previously named "Memory Optimizer" Variable Graphics Memory Advisory for 075 and for 080
NVMe RAID 0 & RAID 1*	Disable	Enable	Enable	*Storage password in NVMe RAID mode not supported. Hot swapping is not recommended for disks that are part of AMD-RAID boot array.
Mobile Overclocking - Precision Boost Overdrive (PBO)	Disable	Enable	Enable	
Mobile Overclocking - Advanced Precision Boost Overdrive	Disable	Enable	Enable	
Max Supported Memory Speeds – LPDDR5	Enable	Enable	Enable	
AIM-T (AMD Integrated Manageability Technology)	Disable	Disable*	Enable	*Issue listed in 080 Advisory. To be fixed in 1.0.0

STRIX HALO OPERATING SYSTEMS SUPPORT



Operating System	Alignment	Comments
Windows 11 23H2	Supported starting with SW Stack 1.0.1	64-bit only
Windows 11 24H2	Aligned to production release of drivers	64-bit only
Ubuntu Linux	Ubuntu LTS 24.04 with appropriate OEM kernel	64-bit only
RHEL Linux	RHEL 10.x	64-bit only

Notes:

- No Windows7 / Windows 8/ Windows 8.1 Support/Windows 10 22H2
- No support for 32-bit UEFI firmware
- Graphics driver alignment is subject to change
- No support for Legacy BIOS/CSM
- Support for Ubuntu LTS 24.04 based on OEM kernel (not LTS/generic or HWE kernels)
- Support for RHEL contingent upon official OEM request
- Supported features in Linux always submitted upstream to benefit future community Linux distros (Fedora, Debian etc.)

The slide features two large, overlapping geometric shapes. A teal shape forms a large triangle on the left and bottom, while an orange shape is positioned above it, also in a triangular form. The text is centered within the teal area.

Linux OS Feature Enablement

STRIX HALO FEATURE ENABLEMENT¹ – LINUX OS



	Supported	Comment
APU GFX (UMA) kernel support	Yes	
USB DP Alt-Mode	Yes	DP1.4a Alt Mode
USB 2.0 / 3.2 (5Gbps) / 3.2 (10Gbps)	Yes	
USB 4.0 CM core functionality	Yes	
SD Card 7.0 (External)	Yes	
Standby (S3)	No	
Cold Boot (S5)	Yes	
S2IDLE (S0i3)	Yes	
USB 3.2 Gen 2 (10Gbps)	Yes	
S4 (Hibernate)	No Yes	Incompatible with UEFI Secure boot. With condition to disable UEFI Secure Boot, as S4 is Incompatible with UEFI Secure boot
I2C / SMBUS, UART, GPIO, WDT, UCSI	Yes	
AMD Wireless Button Driver	Yes	
IOMMU	Yes	
PCIe (x1, x4, x8) Gen 1 to Gen 4	Yes	
DisplayPort 1.4a	Yes	
4K at 24, 30 and 60 FPS video	Yes	
MIPI Camera	Yes ¹	Ubuntu at launch RHEL will be supported later
Sound wire audio	Yes ¹	Limited codec support

¹Contact your customer representative regarding status of features prior to these release dates or features not listed.

STRIX HALO FEATURE ENABLEMENT¹ – LINUX OS



	Supported	Comment
AMD PMF Smart PC Solution Builder	Yes	

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¹Contact your customer representative regarding status of features prior to these release dates or features not listed.

The background features abstract geometric shapes. A large teal shape occupies the bottom half, while an orange shape is positioned in the upper left. The text "Software Enablement" is centered within the teal area.

Software Enablement

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STRIX HALO AUDIO CO-PROCESSOR SYSTEM FEATURES



Feature Name	Description	Benefits
TinyNNE Addition (SW support of framework)	Added Tiny NNE engine. Support TensorFlow Lite model deployment to Tiny-NNE	Improve battery life on speech and audio inference use cases. Potential lead over Intel, parity with QC/Apple
High performance premium low-power AI noise reduction on Tiny-NNE	Use Tiny-NNE to offload best-in-class AI noise reduction at reduced power. Includes required front-end processing (AEC, BF, AGC as required for Teams cert).	Improved system power savings in Teams with ACP
Improved ACP SDK and tools	Improve ACP SDK and tools for better rapid deployment and reconfiguration for OEMs	Faster time-to-deployment for ACP features
AMD noise reduction and AEC algorithm deployment	Provide a solid, Teams-certified AI-NR and AEC solution to OEMs at no cost	Higher baseline of Teams audio performance and power savings across the ecosystem with AMD

Decorative geometric shapes in orange and teal colors, consisting of several overlapping trapezoids and parallelograms.

Graphics, Audio, Display Interfaces ▲

GRAPHICS & VIDEO



Interfaces	Strix Halo
Graphics Core	Radeon Navi3.5 Core Architecture Variable Rate Shading (VRS) Ray Tracing Acceleration Mesh Shaders 32b HDR Format (999e5) Filter, Blend, Display ML/Inference Mixed Precision Ops (DLOps) Corner Sampling (PTEx) Lens Matching Rasterization(FSR) Stereo (Double Position) Shader Post depth access Compressed UAV/ROV Shader Access Dirty Rectangle Detect Improved Compute Dispatch PS with only UAV access
DirectX	12
APU/dGPU Smart Shift 2.0	No
VCN / Video Controller Next	2 instance of VCN4.0 AOMedia (AV1) encoder & decoder VP9, H.264, H.265 decode with 8bit and 10bit depth and upto 4K@24, 30 and 60 FPS video at Vmin H.264, H.265 8bit video encode up to 4K@30 or 60 fps H.264 B-frame Encode format conversion (tone map, color gamut) RGB reader for encode 8K encode (HEVC) Improved video encoder quality

DISPLAY FEATURE OVERVIEW



Strix Halo				
Feature		Internal Display (eDP)	External Display - DP/DP alt-mode	External Display - HDMI
FreeSync Dynamic Refresh Rate		Yes	Yes	Yes
Dynamic Contrast		Yes	Yes	Yes
FreeSync Premium Pro – PQ HDR		Yes	Yes	Yes
FreeSync Replay - support IPX (STX)		Yes	N/A	N/A
Seamless Pixel Rate Switch (SPRS)		Yes	N/A	N/A
VariBright 2.0 Support (LCD panel type)		Yes	N/A	N/A
Varibright xLED (OLED/miniLED)	Smart Color Profile (SCP)	Yes (OLED Only)	N/A	N/A
	OLED Edge Diming	Yes	N/A	N/A
	Content-Adaptive Contrast Profile (CACP)	Yes	N/A	N/A
Dynamic Refresh Rate Control (DRR; Win11 Feature) v2		Yes	Yes	N/A
Panel Self Refresh	PSR1	Yes	N/A	N/A
High Dynamic Range 10 bit (HDR10)		Yes	Yes	Yes
Brightness 3 on SDR Panel		Yes	N/A	N/A
Panel Calibration		Yes	Yes	Yes

STRIX HALO DISPLAY INTERFACES



Interfaces	Strix Halo
Display Outputs	Five (Two USB-C w/USB4 support; One USB-C w/USB3.2 support) Each 1x4 capable of DP, HDMI or DVI Hot plug support Up to 4 Simultaneous Display
USB-C / DP Alt.Mode	Two ports for USB4 or USB-C with /DP MUX, One port USB3.0/DP MUX on USB-C Requires external PD controller DP2.1 alt mode support HDCP 2.3
DisplayPort	DP 2.1 up to 10G DP (HBR3, 8.1 Gbps) at 0.6V DSC 1.2a (FEC is required) HDCP 2.3 (HDCP On Demand) Multi-Stream Transport (MST) LTTPR
Embedded DisplayPort (eDP)	Version 1.5, DSC 1.2a (FEC is required) ASSR (required for Content Protection) Dual eDP (DP1 can be used as eDP with backlit support) Intermediate Link Rate (ILR)
HDMI	HDMI 2.1 FRL PHY with DSC, up to 12G with Re-driver DSC1.2a supported in HDMI 2.1 FRL mode HDMI 2.1 VRR Full HDR support, HDCP 2.3 (HDCP On Demand)
DVI	1920x1200@60Hz max
Wireless Display	Miracast

STRIX HALO HDMI 2.1 FRL & DSC



▲ HDMI Fixed Rate Link (FRL) enables new bandwidths beyond the 18 Gbps limit of TMDS

- FRL uses link training and error correction to ensure signal quality (common problems with TMDS)
- FRL enables transport of DSC compressed video to further improve potential video timings

▲ Strix Halo FRL enables:

- 8K60 at Vmax
- 4K120 at Vmin

▲ DSC enables all Strix Halo use cases at 6 Gbps FRL rate

- DSC is optional for displays, so recommend 10 Gbps to ensure interop with displays without DSC

Color Format	4K60	4K120	8K60
SDR 4:4:4 (24bpp)	6 Gbps	8 Gbps	N/S
HDR 4:4:4 (30bpp)	6 Gbps	10 Gbps	N/S
HDR 4:4:4 (36bpp)	6 Gbps	12 Gbps	N/S
HDR 4:4:4 (DSC)	3 Gbps	6 Gbps	6 Gbps
SDR 4:2:0 (12 bpp)	3 Gbps	6 Gbps	8 Gbps
HDR 4:2:0 (15 bpp)	6 Gbps	6 Gbps	10 Gbps
HDR 4:2:0 (18 bpp)	6 Gbps	6 Gbps	12 Gbps

STRIX HALO APU DISPLAY OUTPUT SUPPORT



Port	Version	Example formats Max resolution W/O DSC	Example formats Max resolution With DSC	Max Link Rate	Re-timer/Re-driver
eDP	1.5	3840*2160@240Hz 8bpc 7680*4320@60Hz 8bpc	3840*2160@240Hz 10bpc 7680*4320@60Hz 10bpc	UHBR10*	Not needed if MBDG trace length guidance is met
DP	2.1	MST:2 x 3840*2160 @60Hz 10bpc	MST:4 x 3840*2160@60Hz	UHBR10	Not needed if MBDG trace length guidance is met
		Single: 3840*2160 @120Hz 10bpc	Single:7680*4320@60Hz		
USB-C DP-Alt	USB 3.2	MST:2 x 3840*2160 @60Hz 10bpc Single: 3840*2160 @120Hz 10bpc	MST:4 x 3840*2160@60Hz Single:7680*4320@60Hz	UHBR10	Not needed if MBDG trace length guidance is met
	USB 4.0	MST:2 x 3840*2160 @120Hz 10bpc Single: 3840*2160 @240Hz 10bpc	MST:4 x 3840*2160@120Hz Single:7680*4320@120Hz	UHBR20	Retimer required
HDMI	2.1	3840*2160@120Hz 10bpc 8K60A 10bpc (4:2:0)	8K60B(4:4:4)	12G	FRL8 ok for FR-4 up to 5" traces FRL10/12 needs redriver

* Depends on UHBR10 eDP panel availability



4K MONITOR DOCKING ON USB-C DISPLAY ALT. MODE

▲ Keeping USB 3.x support to dock active → Therefore 2 lane DP Alt Mode

	4K60 SDR	4K60 HDR	4K120 SDR	4K120 HDR
USB4 – 10G	1 head	N/S	N/S	N/S
USB4 – 10G + DSC	3 heads	3 heads	1 head	1 head
DP 2.1 Alt – 10G	3 heads	3 heads	2 heads	2 heads
USB4 – 20G	2 heads	1 head	1 head	N/S
USB4 – 20G + DSC	3 heads	3 heads	3 heads	3 heads

Decorative geometric shapes are located in the upper left and middle of the slide. There is a large orange parallelogram pointing downwards and to the right, and a smaller orange parallelogram pointing downwards and to the left, positioned to the right of the larger one. Below these is a large teal parallelogram pointing downwards and to the right, which serves as the background for the title text.

Memory Implementation Guidance ▲

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UNIFIED MEMORY CONTROLLER (UMC)

- ▲ UMC interfaces the system via Scalable Data Fabric
- ▲ LPDDR5/x: 16 UMC controls one memory 16-bit channels, total of 256b memory.
- ▲ LPDDR5/x Memory support
 - 8x32 or 4x64 DRAM Down Support
 - Supports x8 or x16 die width
- ▲ Power Management
 - Clock Gating
 - Power Gating
 - Up to 4 Memory P-states
- ▲ UMC Requirements
 - UMC is required to operate at the same MEMCLK frequency, regardless of the channel

VALID MEMORY CONFIGURATIONS AND MAX DATA RATE



DRAM DOWN—LPDDR5/x

Memory Type	APU TDP	Board Layers	DRAM Type	Number of DRAM Populated	DIMM Rank	Max Data Rate
LPDDR5/x	All	12L with u-Via	x32 315 ball (Utilizing x16 or x8 die)	8	LP only	LPDDR5x-8000
LPDDR5/x	All	12L with u-Via	x64 496 ball (Utilizing x16 or x8 die)	4	LP only	LPDDR5x-8000

Two orange geometric shapes: a large parallelogram and a smaller parallelogram to its right, both pointing towards the bottom right.

USB4 Implementation Guidance ▲

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USB4 RETIMER – COMMON FOOTPRINT

Vendor	Part Number	Sample Date
Parade	PS8835	Now
Kandou	KB8015	Q2-24
Analogix	ANX7452	Q1-24
Nuvoton	KM864743	Dec '23
Mediatek	MT9860	TBD, est. Q4
Novatek	TBD	TBD

Common footprint parts are available at Strix Halo time frame, different from Strix

For new motherboard designs, please refer to the STXHL AVL for the full list of USB4 retimers as they become available

USB-PD COMMON FOOTPRINT COMPONENTS



Provider	Part Number	Schedule (ES)	Schedule (MP)	EPR/SPR	Single or Dual Port
IFX	CYPD6129	December 2023	September 2024	SPR	Single
IFX	CYPD6229	December 2023	September 2024	SPR	Dual
ITE	IT5271	Now	April 2024	SPR	Single
ITE	IT5272	Now	April 2024	SPR	Dual
Realtek	RTS5452P	Now	Q4'2023	SPR	Single
Realtek	RTS5453P	Now	Q3'2023	SPR	Dual
TI	TPS66993	November 2023	Mid 2024	SPR	Single
TI	TPS66994	November 2023	Mid 2024	SPR	Dual
Analogix	ANX7415	Now	October 2023	EPR	Dual
ITE	IT5273	Now	May 2024	EPR	Single
ITE	IT5274	Now	May 2024	EPR	Dual
IFX	CYPD8129	March 2024	September 2024	EPR	Single
IFX	CYPD8229	March 2024	September 2024	EPR	Dual
Realtek	RTS5452S	Q1'2024	Q2'2024	EPR	Single
Realtek	RTS5453S	Q1'2024	Q2'2024	EPR	Dual
TI	TPS66993E	November 2023	Mid 2024	EPR	Single
TI	TPS66994E	November 2023	Mid 2024	EPR	Dual

Common footprint parts are available at Strix Halo time frame, different from Strix

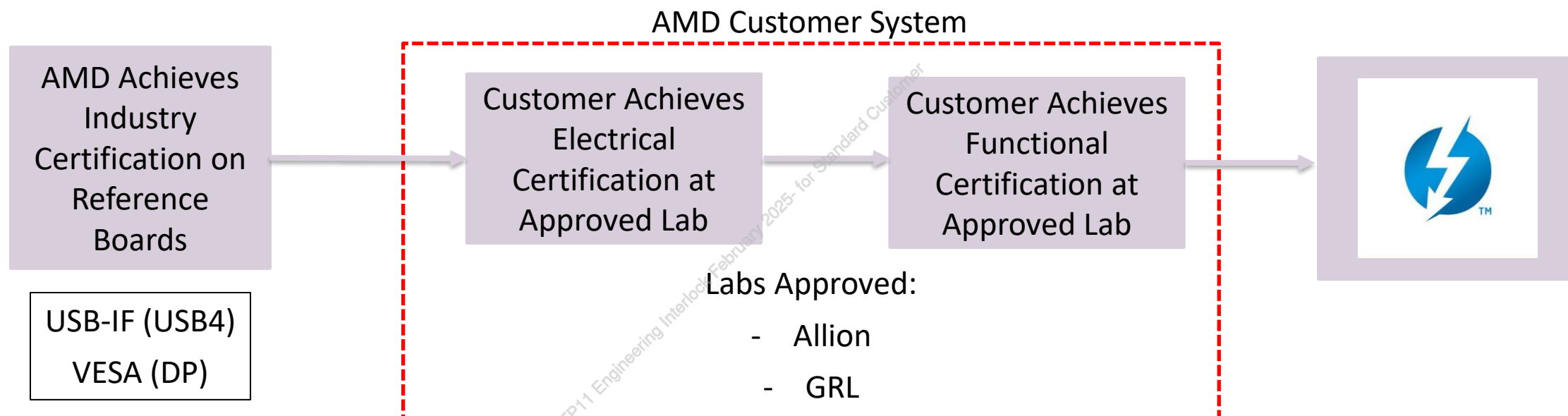
For new motherboard designs, please refer to the STXHL AVL for the full list of USB-PD controllers as they become available

AMD: PATHWAY TO THUNDERBOLT 4 CERTIFICATION



▲ Thunderbolt 4 is a full feature USB4 implementation – AMD systems can qualify for Thunderbolt 4 Branding

▲ Pathway To Thunderbolt 4 Certification and Branding of AMD Strix Based Customer Platforms:



▲ Notes:

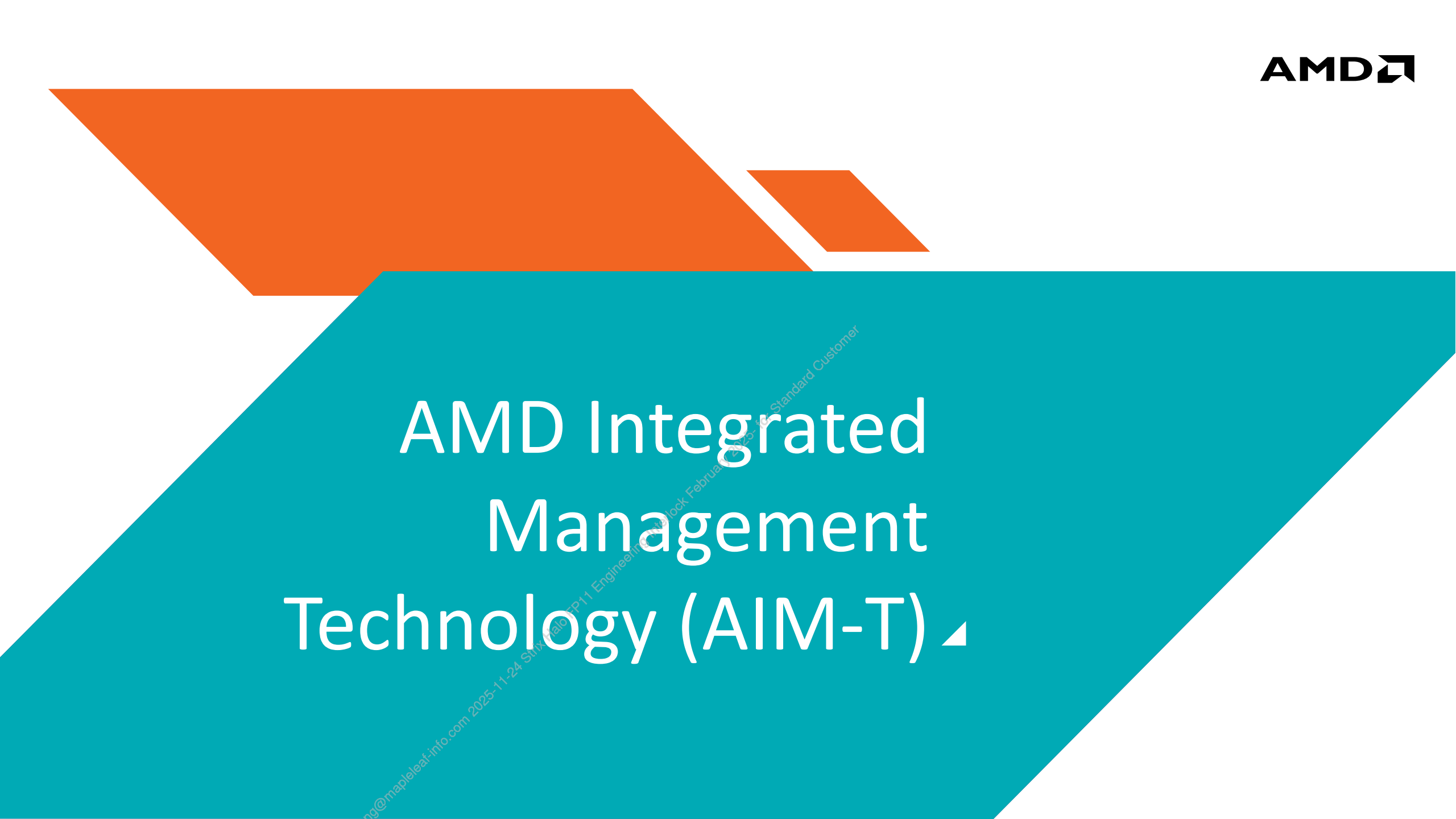
- Requires the customer to submit to 2 locations, where second location does One Day test only
- Options for AMD Customers: 2 independent test labs or single independent lab company with 2 locations

▲ Process is described in Intel Document #639816

STRIX HALO: IT MEETS THUNDERBOLT 4 REQUIREMENTS



	USB4 Minimum Requirement	Thunderbolt 4 Requirement	Strix-Halo
Throughput Speed	20Gb/s	40Gb/s	✓
Tunneled DisplayPort	DP 1.1 (4.32Gb/s)	DP 1.4a HBR3 (34.56Gb/s) 2 Independent Streams (SST)	✓
Tunneled PCIE	Optional	PCIE Gen3x4	✓
Thunderbolt 3 Compatibility	Optional	Required	✓
Native USB speed	USB 3.2 G2 (10Gbps)	USB 3.2 (10Gbps)	✓
System Charging	Optional	Required	Platform dependent
Power Delivery for Devices	7.5W Bus Power	15W Bus Power	Platform dependent
USB4 Specification	Compliant	Compliant	✓
DMA Based Protection	Option	Required	✓
USB-PD Revision	PD 3.1	PD 3.1 (Fast Role Swap also recommended)	AVL adherence
External Graphics Support	No	Yes (Windows)	✓
Pre-OS boot From PCIE	No	Recommended	✓

The slide features a large teal diagonal shape on the left and bottom, and an orange diagonal shape on the top left. The title text is centered in white. A diagonal watermark is visible across the teal area.

AMD Integrated Management Technology (AIM-T) ▲

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AMD INTEGRATED MANAGEMENT TECHNOLOGY (AIM-T) SUPPORT

OEM platform built with	AIM-T Solution from AMD (Qualcomm Hastings Prime WLAN/Hamilton, MTK RZ616/RZ717)	Wired NIC partner Solution (RTK)
Wired Only interface	Assist wired manageability for KVM profile	Provides Wired Manageability
Wireless Only interface	Provides wireless manageability	N/A
Wired + Wireless interfaces	1) Assist wired manageability for KVM profile 2) Provides wireless manageability	Provides Wired Manageability

▲ Strix Halo AIM-T Solution will support wireless manageability and KVM assist for Wired Manageability

POWER STATES IN WHICH WIRELESS MANAGEABILITY WILL BE SUPPORTED

ACPI Global Power States	ACPI Sleep Power States	AMD Power States	Source of Power	Strix Halo Wireless Manageability support planned (Y/N)
G0	S0	AMD Low Power State	Does not matter	Y
G0	S0 low-power idle	S0 low-power idle (Modern Standby)	AC Power	Y* (Will support this through the OS wakeup [DASH Pattern Match])
G0	S0 low-power idle	S0 low-power idle (Modern Standby)	Battery	N
G1	S3	S3	Does not matter	N
G1/G2	S4/S5	S4/S5	AC Power - Referred to as 'G1(AC)'/ 'G2(AC)'	Y
G1/G2	S4/S5	S4/S5	Battery - Referred to as 'G1(AC)'/ 'G2(AC)'	N
G3	Mechanical Off	Mechanical Off		N

Software Tools

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SOFTWARE TOOLS - AMD SPECIALIZED TOOL SUITE



Tool	Purpose	Initial Release Date
APU Validation Toolkit (AVT)	An integrated toolkit that covers power and thermal management functionality. See more detailed feature list on next slide.	SW Stack 0.7.0
System Stress Test Tool (SST)	Used by validation labs to verify stability of AMD processor-based computing systems • Concurrently runs specialized diagnostics that target system components • Tests may include CPU core, cache, memory, storage USB/eMMC, PCIe, GPU compute, graphics, video, network and other system level test modules. • Used to find problems such as hardware logic errors, bus crosstalk, underpowered signal drivers, and software bugs in device drivers	SW Stack 0.7.0

APU VALIDATION TOOLKIT (AVT)



Sub Features	Purpose	More Details		
PStates	Used to test performance states on systems using AMD processors	- Core p-state - NB p-state		- Memory p-state - P-state table manipulation
PMM	Allows viewing/configuration of PMM features of processor such as:	- Core Effective frequency - Max-Freq, Min_Freq, OS Power Schemes		- SVI2,SVI3 features - Monitor, ramp up/down CPU utilization - Device States (Windows Only)
CxTest	Tests APU C-State feature	Force processor into various C-States, Inhibit C-states (0% to100%), Monitor C-State residency		
Thermal Charz Tool (TCT)	Thermal testing and characterization	Thermal testing – Runs power virus patterns on user-selected cores		
Thermal Checker	Monitor and configure thermal control functions of systems	SB TSI temperature, PROCHOT status, ACPI Thermal Zones		
Performance Counters	Allows to select and monitor various performance events	Events for – Core, Data Fabric, FCH		
System Access Utility (SAU)	Read/write to various registers in system	- ACPI Tables - MSR (x86) - Memory	- IO/MMIO - PCI - SMBus	- CPUID - Debugger console for – PCI, MSR, IO/MMIO, Memory, SMBus, CPUID - Python Scripting support for above register access.
System Information	Widget displaying System information	- Processor – Name, Family-Model-Stepping, cache size etc.. - BIOS Details, Memory Devices - Size, speed, type, form factor, etc		
Logger	Allows consolidated logging. Can be launched from command line for background logging.			
Cmd line support				

SOFTWARE TOOLS - OTHERS



Tool	Purpose	Initial Release Date
AMD SystemDeck	Tool to verify SoC Power management feature working for Windows/Linux	SW Stack 0.7.0
AMD Register Explorer (RegXP) -> AMD Register Accessor ARX	Register R/W Debugger	SW Stack 0.7.0
AMDPAT	Platform characterization and bus testing tool used to check memory channels	SW Stack 0.7.0
MEMEYE (1D Support)	DDR Margining Tool for memory margin	SW Stack 0.7.0
AMD XIO	Eye Diagram PCIe/USB Interfaces	SW Stack 0.7.0

NDA Documents

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STRIX HALO SYSTEM DESIGN DOCUMENTS – [DEVHUB.AMD.COM](https://devhub.amd.com)



Design Guides and Design Files	Part Number	Revision	Release Date
Functional Data Sheet for FP11 Processors	58173	0.70	Sep'23
Motherboard Design Guide for FP11 Processors	58235	0.70	Dec'23
Infrastructure Road Map (IRM) for FP11 Processors	58023	1.11	Aug'24
Flotherm Thermal Model User Guide for FP11 Package Processors	58202	0.60	Aug'23
Thermal Design Guide for FP11 Processors	58201	0.60	Junr'23
Motherboard Schematic Checklist for FP11 Processors	58233	0.50	Jun'23
Motherboard Layout checklist for FP11 Processors	58234	0.50	Jul'23
NOVA Signal Conditioner Self-Certification Testing Document	56650	1.06	Nov'23
NOVA Daughtercard Design Guide	56717	1.03	Jul '23
BIOS Porting Guide for Family 1Ah Models 70h-77h	58405	0.30	Feb'24
AMD BIOS Design Guide for Microsoft Pluton Security Processor Enablement	57174	1.20	Dec'22
Processor Programming Reference (PPR) for AMD Family 1Ah Models 70h , Revision A0 Processors	57930	1.54	Oct'23
Power and Thermal Datasheet for AMD Family 1Ah Models 70h-77h Processors	58242	0.50	Mar'25
AMD Approved Vendor List (AVL) for System components for AMD Family 1Ah Models 70h-77h Processors	58240	0.72	Aug'24
Modern Standby BIOS Implementation Guide	56358	1.06	Jan'24
Platform Performance and Power Optimization Guide for AMD Family 1Ah Models 70h-77h FP11 Processors	58237	0.50	Mar'25

STRIX HALO SYSTEM DESIGN DOCUMENTS – [DEVHUB.AMD.COM](https://devhub.amd.com)



Design Guides and Design Files	Part Number	Revision	Release Date
Electrical Data Sheet(EDS) for AMD Family 1Ah Models 70h-77h Processors	58241	0.50	Jul'23
Sensor Fusion Hub Design Guidance for FP11 Processors	58613	0.50	NA
Processor Sensor Fusion Hub (SFH) BIOS Implementation Guide	56371	1.00	Aug'18
Modern Standby BIOS Implementation Guide	56358	1.06	Jan'24
AMD FreeSync™ Specification	56947	4.20	Jan'24
AMD System Peripheral Bus Overview (NDA)	55861	1.04	Nov'22
AMD Serial VID Interface Version 3 (SVI3) Specification	56413	3.00	Feb'24
Maple Functionality Guide for FP11 Processor Platform	58230	1.00	Mar'24
Maple User Guide for FP11 Processor Platform	58231	1.00	Apr'24
Strix Halo Design Platform EC Module Interface Specification for Family 1Ah Models 70h-77h	56983	1.00	Dec'23
AMD Mobile Ergonomic Specifications	57622	0.60	May'22
AMD 2-bit VID™ Converter Device Guidelines	57646	0.60	May'23
AMD Platform Security Processor BIOS Architecture Design Guide for AMD Client Processor (NDA)	55758	2.03	Nov'23
AMD Generic Encapsulated Software Architecture (AGESA) V9 Interface Specification (NDA)	55483	1.72	Mar'23
Secure Biometrics BIOS Implementation Guide	56560	1.01	Jun'23
System Temperature Tracking Tool (STT) and User Guide for FP11 Strix Halo Processors	57379	1.21	Aug'24

STRIX HALO SYSTEM DESIGN DOCUMENTS – [DEVHUB.AMD.COM](https://devhub.amd.com)



Design Guides and Design Files	Part Number	Revision	Release Date
AMD Solder Reflow Profile Guide	55592	1.07	Dec'22
AMD I/O Virtualization Technology (IOMMU) Specification	48882	3.08	Feb'24
Acoustic Design Guideline of Ensuring Microphone Performance	57650	1.30	Nov'22
AMD System Validation Manual Addendum (NDA)	58524	0.50	Mar'24
AIM-T 1.0 Solution Integration Guide	57061	1.10	Mar'23
AMD USB Power Delivery I2C Target Programming Guide for APU Gen2.4	57196	0.50	May'23
AMD Power Delivery I2C Target Programming Guide for APU Gen2.3	57239	0.53	May'23
Validation Platforms PD Module Interface Specification	57890	0.90	Dec'23
SOLP Low Power State Enablement Guidance(Place Holder)	NA	0.50	NA

Revision History ▲

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REVISION HISTORY



Revision	Release Date	Changes
February 2024	2/29/2024	• 1st Engineering Interlock release
April 2024	4/30/2024	• 2nd Engineering Interlock release
June 2024 release	6/13/2024	• 3rd Engineering Interlock release
September 2024 release	9/27/2024	• 4th Engineering Interlock release
February 2025 release	2/21/2025	• 5th Engineering Interlock release (Updates for 1.0.0 and Post 1.0.0)

CHANGE LOG – SINCE SEPTEMBER 2024 RELEASE



Change	Comments
New name “Variable Graphics Memory”	Previously named “Memory Optimizer” See Variable Graphics Memory Advisory for 075 and for 080
PSP Firmware Anti-Rollback (FAR)	Issue listed in 080 Advisory. Fixed in 1.0.0.
AIM-T (AMD Integrated Manageability Technology)	Issue listed in 080 Advisory. Fixed in 1.0.0.
eDP ASSR UHBR10 scrambling support UHBR10 for eDP	No longer POR. Post 1.0.0 proof of concept.
Display Port 2.0 (UHBR10)	Fixed in 1.0.0
AMD Approved Vendor List (AVL)	Updated in August 2024
S4 Hibernate on Linux	Conditionally supported (page 34)
Added Strix Halo system diagram on page 9	
Document Updates pages 61-63	Updated documents in blue text.
Win 11 23H2	Supported with SW Stack 1.0.1

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